ISM6642 — Lecture 1

Machine Learning for Business Applications

Review & Business Statistics Introduction

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DISCUSSION TOPIC

Can Machines Acquire Knowledge?

How does a child learn to recognize a cat after seeing a few examples?

How does Google Photos recognize the same child across 10,000 photos?

FOUNDATIONS

What is Machine Learning?

An essential component of **artificial intelligence** that uses algorithms to examine data and automate the creation of analytical models.

Computers discover patterns in data **without explicit programming** — a major advantage in today's data-driven business landscape.

Applied across **marketing, supply chain, CRM,** and financial forecasting.

• PRACTICAL EXAMPLE

Netflix Recommendations

80% of content watched on Netflix comes from its ML recommendation engine — not user searches. It learns your taste from billions of viewing signals.

Spotify Discover Weekly

ML analyzes 30M+ songs and your listening history to curate a personalized playlist every Monday — no explicit rules written.

TOOLS

Software for Machine Learning

Python	R	MatLab	Spark	Julia
Industry Leader NumPy · Pandas Sklearn · TensorFlow	Statistics & data visualization	Engineering & matrix operations	Distributed large-scale processing	High-performance numerical computing

- **Quick start:** Anaconda distributes Python with all major ML libraries pre-installed — anaconda.com/download. Used by Google, Facebook, and NASA for production data science.

CONTEXT

Machine Learning vs. Automation

3RD INDUSTRIAL REVOLUTION

Automation

Computers execute **known, fixed rules**. Digitization of existing processes.

Payroll processing, invoicing, regional sales summarization

4TH INDUSTRIAL REVOLUTION

Machine Learning

Computers **learn and develop intelligence**. Rules are discovered from data.

Loan approval, fraud detection, price optimization

• CASE STUDY

JPMorgan COIN

Reviews **12,000 commercial credit agreements** in seconds.

Previously needed **360,000 lawyer-hours** annually.

ML discovered what to look for — no rules were programmed explicitly.

FUNDAMENTALS

What is Data?

Data

Numbers or text collected through a **measurement process**

72°F — a temperature reading from a sensor

1,523 — clicks on a product page Monday

$4.99 — a credit card transaction at 3:14 PM

Information

Result of **analyzing data** to extract meaning that supports evaluation and decision-making

"Site traffic peaks Tuesday mornings — schedule campaigns then"

"Region 3 customers spend 30% more on average"

FUNDAMENTALS

What is Data Analytics?

The use of **data, information technology, statistical analysis,** and mathematical models to help managers gain better insight and make **fact-based decisions.**

Business Analytics (BA) is a subset of Data Analytics — specifically focused on business operations and decisions.

• PRACTICAL EXAMPLE

Amazon

Uses analytics to **predict what you'll buy before you even search** — their recommendation engine drives ~35% of total revenue.

Also drives warehouse stocking, same-day delivery routing, and dynamic pricing — all data-informed decisions.

FRAMEWORK

Types of Data Analytics

Descriptive

What happened?

→ Evaluates past and current performance

→ Organizes data into charts and reports

→ Summarizes information for insight

e.g. Monthly sales dashboard, customer churn report

Predictive

What will happen?

→ Projects future results from historical data

→ Recognizes patterns for future scenarios

→ Anticipates risks and identifies trends

e.g. Demand forecasting, credit risk scoring

Prescriptive

What should we do?

→ Applies optimization for decision-making

→ Minimizes risks or maximizes revenue

→ Recommends best course of action

e.g. Dynamic pricing, supply chain optimization

CASE STUDY

Retail Markdown Decisions

Business Problem: Department stores manage seasonal stock via price reductions.

When is the optimal time to lower prices, and by how much, to maximize revenue?

Descriptive

Analyse historical data on similar items — pricing, units sold, marketing, seasonality.

"Last fall, 40% of coats sold in week 3 of October at full price."

Predictive

Forecast future sales influenced by pricing tactics and external factors like weather.

"A 20% markdown in week 2 of October will clear 60% of inventory."

Prescriptive

Determine the most effective pricing and advertising strategy to enhance revenue.

"Mark 15% off in week 1, 30% in week 3, with loyalty email targeting."

DATA IN THE WILD

Examples of Data Sources and Uses

Traditional Business Data

- Annual reports
- Accounting audits
- Financial analysis
- Economic trends
- Marketing research
- HR measurements

Digital & Behavioral Data

Page views, visitor country, session duration, referral source

Search queries, products viewed, cart abandonment

Purchases, reviews read, social sharing

→ Amazon tracks 200+ behavioral signals per visitor to personalize every page in real-time

SCALE

Big Data — The 4 Vs

V1
Volume

Massive amounts from many sources

Facebook: 4 petabytes/day

V2
Variety

Text, images, video, sensors, transactions

Twitter: text + images + location + video

V3
Velocity

Available and processed in real time

Google: 8.5B searches processed daily

V4
Veracity

Uncertain or unpredictable quality

Social media: noise, bots, sampling bias

> *The effective use of big data has the potential to transform economies... Big data will become a key basis of competition for existing companies, and will create new competitors who are able to attract employees that have the critical skills for a big data world.* ***— McKinsey Global Institute, 2011***

DATA QUALITY

Reliability & Validity

Reliability

Data are **accurate and consistent** — the same measurement yields the same result repeatedly.

Validity

Data **correctly measures what it's supposed to** — the metric captures the intended concept.

Examples

	Reliable	Valid
Tire pressure gauge reading consistently low	✗	✓
Call volume as customer dissatisfaction proxy	✓	✗
Food taste rating for restaurant quality	✗	✗

MODELING

Decision Models

A **logical or mathematical representation** of a problem or business situation that can be used to understand, analyze, or facilitate making a decision.

Can be simple (a scoring formula) or complex (a neural network with millions of parameters)

• PRACTICAL EXAMPLE

FICO Credit Score

A decision model combining payment history (35%), credit utilization (30%), credit history length (15%), and other factors into a single score.

Used by banks to approve or deny millions of loan applications daily — a consistent, auditable model replacing subjective judgment.

CHALLENGE

Opportunistic Data

Operational data is typically collected for a purpose other than analysis.

Multiple business units produce **silo-based data systems** — inconsistent formats, definitions, and ownership.

This makes business analytics **uniquely challenging** compared to controlled experimental statistics.

• REAL-WORLD CHALLENGE

Hospital EHR Data

Electronic health records are collected for **patient treatment**, not research. The same condition may be coded differently across departments, hospitals, and physicians.

Result: aggregating data for analytics requires enormous data-cleaning effort before any modeling begins.

The Weather Channel Knows When You'll Buy Bug Spray

Wall Street Journal — Video Case Study

How a weather data company became a behavioral prediction machine — correlating weather patterns with consumer purchasing decisions to sell targeted advertising to retailers.

DISCUSSION

The Evolving Landscape of Statistics

Discussion Prompt

Explain a data system you engage with that produces a significant volume of information.

In what ways can this data be used for informed decision-making?

What's Changed

- → Vast data collections at unprecedented scale
- → Dramatic drops in storage and processing costs
- → ML tools that are now practical and accessible
- → Non-linear models and multi-stage decision strategies now feasible

COMPARISON

Traditional Statistics vs. Machine Learning

Traditional Statistics

1. Develop a hypothesis
2. Collect relevant data
3. Test hypothesis, draw conclusions

e.g. Does price affect demand? Collect survey data, run a regression, test statistical significance.

Machine Learning

1. Collect large amounts of data
2. Try different models on the data
3. Find patterns or build a predictive tool

e.g. Gmail spam filter — no rules were written for spam. ML learned from millions of labeled emails.

TAXONOMY

Types of Machine Learning

Unsupervised Learning

Identifies patterns with **no labeled data**.

→ Target segments shoppers into groups — no one defined the groups in advance.

Supervised Learning

Predicts values or categories from **labeled training data**.

→ Gmail spam filter trained on millions of emails labeled "spam" / "not spam".

Semi-Supervised

Only a **portion of data is labeled** for the target value.

→ Google Photos: label a few faces, it identifies that person across thousands of photos.

Reinforcement Learning

Learns optimal decisions via **reward and penalty** signals over multiple stages.

→ Tesla Autopilot and AlphaGo: learned by playing millions of iterations, not from labeled examples.

IMPACT

Applications of Machine Learning

Credit
Scoring, loan approval, default prediction

FICO used in 90% of US lending decisions

Customers
Segmentation, churn prediction, personalization

Starbucks predicts churn 2 months in advance

Fraud
Detection of fraudulent activities in real time

PayPal flags fraud across $1.3T in transactions

Finance
Portfolio management, private equity, AML

Renaissance Medallion: ML-driven, 66% avg annual return

Translation
Language translation across text and speech

Google Translate: 100B words/day, 133 languages

Voice
Recognition of voice and natural language

Siri, Alexa, Google Assistant — billions of queries

Healthcare
Diagnosis, drug discovery, treatment planning

DeepMind AlphaFold solved protein folding in 2021

Logistics
Route optimization, demand forecasting, inventory

UPS ORION saves 10M gallons of fuel per year

METHODOLOGY

Data Partitioning

Training Set ~60% — In Sample	Validation ~20%	Test Set ~20%

Training Set

Build and fit the model. The algorithm learns patterns from this data.

→ Like studying your textbook before an exam.

Validation Set

Tune model complexity. Stop when validation performance starts to drop.

→ Like practice quizzes to adjust your study strategy.

Test Set

Final out-of-sample assessment. Used once — a true measure of real-world performance.

→ Like the actual exam: unseen questions, one attempt.

LECTURE 1 SUMMARY

Key Takeaways

01 ML enables computers to discover patterns **without explicit programming** — powered by data, not hand-coded rules.

02 Analytics spans **descriptive → predictive → prescriptive** — from understanding the past to optimizing the future.

03 Big Data's **4 Vs** (Volume, Variety, Velocity, Veracity) define the scale — and the challenge — of modern business analytics.

04 **Data quality matters** — reliability and validity determine whether a model learns signal or noise.